

**Kathmandu University School of Education**

**Department of STEAM Education**

**Bachelor of Technical Education (B Tech Ed) Program**

**Information Technology (IT)**

**In Association with School of Engineering, KU**

**Course Curriculum**

**Semester I**

**2022**

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## **Overview of the B Tech Ed Program**

Technical and Vocational Education and Training (TVET) has been one of the prominent sectors globally for a long. Dewey (1916) saw TVET as a tool for education reform in modernizing society. This has been equally vibrant in the present context as scholars (e.g., Maclean & Wilson, 2009) argue that TVET is an education and training that prepares an individual for gainful employment. TVET programs equip young people with the skills, knowledge, and competencies required to enter a particular type of professional career (OECD 2017). Hence, the role of TVET in preparing skilled human resources and enabling them to transition into the career path for gainful employment in specific and the country's economic development in a broad is inevitable. The formal TVET began in Nepal only around mid of the 20<sup>th</sup> century. Nevertheless, occupational skills have been transformed from generation to generation for a long (Ministry of Education [MoE], 2012). The establishment of the Council for Technical Education and Vocational Training (CTEVT), the governing body of TVET in 1989, was a substantive effort toward its development and expansion (CTEVT, 2019). Additionally, at present, the Centre for Education and Human Resource Development (CEHRD) offering technical diplomas in 284 technical stream schools throughout the country (GoN, 2018) and universities (such as Kathmandu University, Manmohan Technical University, etc.) run bachelor and master in TVET and other 11 federal ministries offer TVET related formal and non-formal programs (MoEST, 2019). The CTEVT, which is in charge of formulating TVET-relevant policies, developing standards for programs and curriculums, coordination, accreditation, monitoring, and supervision (CTEVT, 2019) largely shares the TVET related activities. There are 1131 CTEVT affiliated and constituent institutions with roughly the average capacity of 70,000 per annual operate formal TVET programs (CTEVT, 2020). Further expansion of TVET institutions in the country is yet to be expected soon since the government has a provision of establishing at least one technical school in all local levels.

Bachelor of Technical Education is a program to develop instructors and teachers in the field of Technical and Vocational Education and Training (TVET) with a view to fulfil the gap of acute shortage of instructors and TEVT professionals in the country which aspires to promote technical and vocational education two-third of the educational system and processes. The program has envisaged that a mix of two or more of the work-based learning, activity-based instruction, on the job training, project-based learning and problem-based learning be used to deliver the program. Because of the 'practical focus' of the program, the assessment has been proposed to address the needs of experiential learning.

The 136-credit program has been structured four types of the courses, such as Educational Core (18), Vocational Specialization (80), Vocational Pedagogy (20), and Educational Research and Internship (18) with a view to inculcate both pedagogical and vocational skills. The vocational specializations have been offered in a manner that they can be utilize hands-on activities as well. The program begins with the foundational courses in general education, vocational pedagogy and the area of specialization. In the later part of the program, OJT/WBL and forms of experiential learning shall be introduced. The culmination of the program will be internships in their specializations and a research report that addresses the problem of vocational education.

### **B Tech Ed in Information Technology**

Bachelor of Technical Education is a 4-year (8 semesters) program to develop instructors, teachers, professionals in the field of Technical and Vocational Education and Training (TVET) with a view to fulfil the gap of acute shortage of instructors and TEVT professionals in the country which aspires to promote technical and vocational education two-third of the educational system and processes. The program has envisaged that a mix of two or more of the work-based learning, activity-based instruction, on the job training, project-based learning and problem-based learning be used to deliver the program. Because of the ‘practical focus’ of the program, the assessment has been proposed to address the needs of experiential learning. The B Tech Ed in IT program aims at providing technical, vocational, and educational exposures for students who can become IT professionals, IT Instructors, TVET educators, TVET curriculum developers, etc. The occupational/specialization courses emphasize the development of core IT skills in the field of computer science and technology innovation. The B Tech Ed in IT graduates will be eligible for doing master's program in IT, TVET, Education, Social Sciences, etc.

### **Major Objectives of the B Tech Ed in IT Program**

- Demonstrate Level 6 competencies envisaged by the National Vocational Qualification Framework (NVQF) of Nepal,
- Exhibit comprehensive, meaningful, and coherent knowledge and skills in the field of technology and computer science
- Apply occupational specialization knowledge and skills in solving problems occurred in their contexts of work,
- Apply vocational pedagogical knowledge and skills in technical and vocational instruction of Information Technology,
- Show scholarly literacy, communication, quantitative reasoning, critical thinking, learning skills needed for advanced and work-based learning,
- Exemplify a deep and principled understanding of the technical and vocational learning processes and the role of the instructors in facilitating these processes in the students,
- Apply a wide range of teaching process skills including curriculum development, lesson planning, materials development, assessment, and pedagogical approaches, and
- Enable students to become resilient and lifelong learners by harnessing internal and external assets as TVET professionals.

### **Career Prospects**

- TVET Instructors and Educators
- TVET Curriculum Developers
- IT Professionals (Software and Apps developer, programmer, Network Engineer, Web Developer, IT Admin, IT Engineer, etc.)
- Eligible for doing master's program in IT, TVET, Education, Social Sciences, etc.

### **Entry Requirements**

The minimum prerequisite for this program is the completion of Level 5 education (as per National Vocational Qualification Framework) in relevant vocational specializations. The details are-

- Minimum D+ grade in each subject of grade 12 with CGPA 1.8 or more  
Or
- A minimum score of second division marks in 10+2, PCL or equivalent in any disciplines

## Core Courses

### EDUC 100: The Teaching Profession

**Course:** The Teaching Profession

**Code:** EDUC 100

**Credit Hours:** 3 (2 Th + 1 PR)

**Nature:** Theory and Practical

#### Course Description

Teaching is a noble profession through which teachers facilitate students' learning to construct knowledge and develop skills to meet the goals of education. A truly amazing teacher is hard to find, difficult to part with, and impossible to forget. A teacher inspires and empowers students through the proper use of 'Hand, Heart and Mind'. For this, a teacher needs to develop professional knowledge and skills through experience and formal education. One of the essential features among teachers is that they should be acquainted with different philosophical orientations of teaching professions to develop their professions as per the demanding need of the local and global contexts and develop their better version. In this context, this course is designed to illustrate the various evolution of the teaching profession in different cultures and civilizations. In this, students will be acquainted with the professional standards of the teaching profession to be better teachers.

Similarly, students will develop a sound understanding of teaching as mentoring and coaching. Moreover, it is necessary to identify and explore different models (reflective practice) of the development of the teaching profession. So, this course will be beneficial for everyone who can see themselves being a professional teacher with knowledge and skills of various teaching professional features.

#### Major Learning Outcomes

- Develop a sound understanding of different philosophical orientations of teaching professions in the local and global contexts
- Exhibit skills of teaching as a process of continuous growth and development of teachers through reflective practice
- Demonstrate the roles of teacher as mentor and coach
- Appraise professional standards of the teaching profession
- Promote 21<sup>st</sup> century core skills and deep learning competencies
- Appraise professional standards of the teaching profession to address the globalized economy
- Develop positive attitudes and dedication towards teaching.

#### Contents in detail with Specific Objectives

| Contents                                                                                                                | Specific Objectives                                                                                                                                                                                                                                                             | Resources                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Unit I: Philosophical Foundation and Evolution of Teaching Profession (8 hrs.)</b>                                   |                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                 |
| <ul style="list-style-type: none"><li>• Historical root of teaching profession</li><li>• Teacher in classical</li></ul> | <ul style="list-style-type: none"><li>• Conceptualize teaching as a noble job and profession</li><li>• Investigate and research on a historical root and foundation of teaching profession</li><li>• Explain and analyze historical roots of teaching as a profession</li></ul> | <ul style="list-style-type: none"><li>• J.R. Young, L.B. Erikson (2011). <i>Imagining, Becoming, and Being a teacher: How Professional history mediates teacher educator identity. Studying Teacher Education</i>, 7 (2), pp. 121-129</li></ul> |

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| <p>and modern times</p> <ul style="list-style-type: none"> <li>Teacher as expert, facilitator and change agent</li> </ul> | <ul style="list-style-type: none"> <li>Analyze and differentiate teacher as a profession in classical and modern times based on their roles, responsibilities, assumptions of society, etc.</li> <li>Compare and contrast the roles and responsibilities of a teacher as expert, facilitator, and agent of change</li> <li>Conceptualize the roles of teacher in teaching profession as an agent of change and design some models of describe facilitator as agent of change in education and society.</li> </ul> | <ul style="list-style-type: none"> <li><a href="https://www.sciencedirect.com/science/article/pii/S1877042816000549">https://www.sciencedirect.com/science/article/pii/S1877042816000549</a></li> <li><a href="https://www.britannica.com/topic/teaching/The-doctrine-of-in-loco-parentis">https://www.britannica.com/topic/teaching/The-doctrine-of-in-loco-parentis</a></li> <li><a href="#">Importance of Teaching Profession   Characteristics of Teaching - Edmonger</a></li> <li><a href="#">Classical vs. Modern Education: The Principal Difference – Angelicum Academy</a></li> <li><a href="https://drive.google.com/file/d/17mXrJ0ZmWpG_yBO49hurShZkAIvNCc5/view?usp=sharing">https://drive.google.com/file/d/17mXrJ0ZmWpG_yBO49hurShZkAIvNCc5/view?usp=sharing</a></li> <li><a href="https://kusoede.edu.np/pluginfile.php/60786/mod_resource/content/1/Teachers%20in%20Classical%20Age.pdf">https://kusoede.edu.np/pluginfile.php/60786/mod_resource/content/1/Teachers%20in%20Classical%20Age.pdf</a></li> <li><a href="https://cft.vanderbilt.edu/guides-sub-pages/blooms-taxonomy/">https://cft.vanderbilt.edu/guides-sub-pages/blooms-taxonomy/</a></li> </ul> |
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## Unit II: Professional Standards for Teachers (20 hrs.)

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| <ul style="list-style-type: none"> <li>Knowing students</li> <li>Knowing contents and methods</li> <li>Plan and implement teaching effectively</li> <li>Developing and maintaining safe learning environment</li> <li>Assessing students and providing</li> </ul> | <ul style="list-style-type: none"> <li>Identify and discover the roles and characteristics of 'students as learners'</li> <li>Interpret the roles and importance of contents of subject matter or curriculum in teaching and learning context</li> <li>Identify various teaching and learning methods/approaches [e.g., Lecture method, Inductive Method, Deductive Method, Discovery method, Laboratory method, Project-based method, inquiry method, Analytic method, synthetic method, Collaborative &amp; Cooperative Learning, Heuristic method, activity-based method, ICT-integrated method, etc.] and illustrate their strengths, weaknesses, and procedures of teaching and learning</li> </ul> | <ul style="list-style-type: none"> <li><a href="https://www.aitsl.edu.au/docs/default-source/national-policy-framework/australian-professional-standards-for-teachers.pdf">https://www.aitsl.edu.au/docs/default-source/national-policy-framework/australian-professional-standards-for-teachers.pdf</a></li> <li><a href="#">A. Reis Monteiro, The Teaching Profession, Present and Future</a></li> <li><a href="https://drive.google.com/file/d/1MkDRCTKH6kM4gEri15GmvTZYY3JVYlZ8/view?usp=sharing">https://drive.google.com/file/d/1MkDRCTKH6kM4gEri15GmvTZYY3JVYlZ8/view?usp=sharing</a></li> <li><a href="https://drive.google.com/file/d/1oCGdfh0AMH6lkbI65f8A_-RKtVXIqSVx/view?usp=sharing">https://drive.google.com/file/d/1oCGdfh0AMH6lkbI65f8A_-RKtVXIqSVx/view?usp=sharing</a></li> <li><a href="https://drive.google.com/file/d/1LiEbUau1M-MN3PsAQ6Iu-Tf_tyT9ppsk/view?usp=sharing">https://drive.google.com/file/d/1LiEbUau1M-MN3PsAQ6Iu-Tf_tyT9ppsk/view?usp=sharing</a></li> <li><a href="https://drive.google.com/file/d/1sptJsvX-R027cffj3LqEhYuUpZ18_vTl/view?usp=sharing">https://drive.google.com/file/d/1sptJsvX-R027cffj3LqEhYuUpZ18_vTl/view?usp=sharing</a></li> </ul> |
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| <p>feedback for better learning</p> <ul style="list-style-type: none"> <li>Continuing professional development learning</li> <li>Working professionally with communities and beyond</li> </ul> | <ul style="list-style-type: none"> <li>Explain the importance of various plans (Yearly, Unit, and Lesson) in teaching and learning</li> <li>Develop plans (Yearly, Unit, and Lesson) and implement one of them in a simulation activity</li> <li>Explain and interpret the importance and characteristics of safe learning environment</li> <li>Describe and present examples of Assessment and illustrate examples to discuss why Assessment is important in teaching and learning</li> <li>Identify the importance of formative feedback (which answer the questions: Where am I trying to go? – Where am I now? – How do I close the gap?) for teacher in the improvement of students' learning and illustrate various means of providing feedback (Sandwich Model etc.)</li> <li>Interpret the meaning and context of continuing professional learning and explain the importance for teachers</li> <li>Demonstrate professional skills of teacher in simulation and presentation activities</li> <li>Explain the roles and responsibilities of teachers' professional skills while working with communities and beyond</li> </ul> | <ul style="list-style-type: none"> <li>Benjamin's Bloom's Revised Taxonomy<br/>(<a href="https://drive.google.com/file/d/1iMAPu4m9JYalGQcH-HeYZPy8ts4vuU20/view?usp=sharing">https://drive.google.com/file/d/1iMAPu4m9JYalGQcH-HeYZPy8ts4vuU20/view?usp=sharing</a>)<br/>(<a href="https://drive.google.com/file/d/1KIJkro4G1QWErNbVLzTDtbvLApbDtF2a/view?usp=sharing">https://drive.google.com/file/d/1KIJkro4G1QWErNbVLzTDtbvLApbDtF2a/view?usp=sharing</a>)</li> </ul> |
| <p align="center"><b>Unit III: Models of Teacher Professional Development (10 hrs.)</b></p>                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <ul style="list-style-type: none"> <li>The action research model</li> <li>The reflective practice/c</li> </ul>                                                                                 | <ul style="list-style-type: none"> <li>Review and examine different model of Teachers' Continuing Professional Development with their major characteristics/features, strengths, and limitations ,<br/>Compare and contrast among four major CPD model.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <ul style="list-style-type: none"> <li><a href="https://drive.google.com/file/d/1jEN6OfM11l16QH7fVGdFzJO8IJieumkz/view?usp=sharing">https://drive.google.com/file/d/1jEN6OfM11l16QH7fVGdFzJO8IJieumkz/view?usp=sharing</a></li> <li><a href="https://kusoede.edu.np/pluginfile.php/60799/mod_resource/content/1/Module%203.pdf">https://kusoede.edu.np/pluginfile.php/60799/mod_resource/content/1/Module%203.pdf</a></li> </ul>                                          |

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| <ul style="list-style-type: none"> <li>• cycle model</li> <li>• The lifelong learning models</li> <li>• The continuing professional development (CPD) framework.</li> </ul> | <ul style="list-style-type: none"> <li>• Reflective Practice/Cycle, Lifelong, and CPD) models</li> <li>• Develop a classroom research and Action plan</li> <li>• Conceptualize professional learning in the lance of CPD frame work</li> </ul> | <ul style="list-style-type: none"> <li>• <a href="https://kusoede.edu.np/pluginfile.php/56511/mod_resource/content/1/21st%20Century%20Skills.pdf">https://kusoede.edu.np/pluginfile.php/56511/mod_resource/content/1/21st%20Century%20Skills.pdf</a></li> <li>• <a href="https://cft.vanderbilt.edu/guides-sub-pages/blooms-taxonomy/">https://cft.vanderbilt.edu/guides-sub-pages/blooms-taxonomy/</a></li> <li>• <a href="http://www.researchgate.net">www.researchgate.net</a></li> </ul> |
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#### Unit IV: Teacher in the 21<sup>st</sup> Century (7 hrs.)

|                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <ul style="list-style-type: none"> <li>• Transmitter and facilitator</li> <li>• Cultural reproducer and change agent</li> <li>• Facilitator of values clarification</li> <li>• Teacher as learner and researcher</li> </ul> | <ul style="list-style-type: none"> <li>• Define and critically examine teachers' various roles as knowledge transmitters Vs. Facilitators, Cultural reproducers and change agents, Learners and Researchers, etc. Enhance Six Core skills for deep learning (Critical thinking and problem solving, Communication and collaboration. Creativity and imagination,. Student leadership and personal development, Global Citizenship, .Digital literacy)</li> <li>• Evaluate the different roles and responsibilities of teachers in the 21<sup>st</sup>-century deep learning based on their major characteristics, strengths and limitations</li> <li>• Demonstrate/Explain the major requirements of knowledge and skills that teachers and educators need in the 21<sup>st</sup>-century</li> </ul> | <ul style="list-style-type: none"> <li>• <a href="https://journals.sagepub.com/doi/full/10.1177/1745499919829214">https://journals.sagepub.com/doi/full/10.1177/1745499919829214</a></li> <li>• <a href="https://www.aeseducation.com/blog/what-are-21st-century-skills">https://www.aeseducation.com/blog/what-are-21st-century-skills</a></li> <li>• <a href="https://britishcounciluk.eu-west.catalog.canvaslms.com/">https://britishcounciluk.eu-west.catalog.canvaslms.com/</a></li> </ul> |
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#### Methodology and Teaching Techniques

The facilitator(s) can use multiple methods or strategies to teaching and learning listed below. Students must be actively engaged in each and every learning activity. The frequent sharing session and reflection activities should be conducted. So, teaching and learning should be based on constructivist model.

- Lecture Method
- Inquiry-based approach
- Inductive & Deductive Methods
- Demonstration approach
- ICT integrated approach
- Collaborative learning (Peer and group learning)
- Discovery and investigation approach
- Project-based learning

### **Major Teaching and Learning Materials**

Materials are essential to communicate knowledge and skills among learners. Rich learning resources and materials must be used during teaching and learning activities. The facilitator(s) is/are suggested to use the following resource materials based upon the nature of the course contents. Also, the facilitator is responsible for developing these learning resources.

- Teacher's Notes and Hands-out
- PPT developed by the facilitator
- Daily used materials
- Chart and cardboard papers
- Laptop/Desktop and projector
- Books or printed materials
- Other materials

### **Evaluation Scheme**

**In-semester Assessment:** 50% (Home Assignments and class work). While doing this assessment, the facilitator(s) is/are recommended to use more than one strategy listed below.

- Attendance (Compulsory, 70%)
- Home Assignments (which may include the following)
- Reflective writings (These tasks encourage students to explore and reflect on the past experiences/current practices related to teaching profession, develop possible strategies and alternatives to improve the practices, be imaginative to portray their understanding/creativity that stimulate students for bringing their vision/s as agents of change)
- Presentations and demonstration (Individual and group presentations on topics related to the contents)
- Regular Practical Tasks (Teaching and Learning simulations)
- Forum Discussion/Collaborative group work
- Monthly written test
- Quizzes
- Project work (Field visit related to teaching profession followed by report writing)
- Viva-voce
- Open book test

**Note:** During every formative assessment task, a continuous and constructive feedback must be provided to improve the learning of students. Students must be given chances for

improving their grades and learning. So, a Continuous Assessment System (CAS) should be followed.

**End-semester Assessment:** 50% (Examination conducted by Kathmandu University)

**Resources:**

- Buchanan, J. (2020). *Challenging the Deprofessionalisation of Teaching and Teachers*. Springer Singapore.
- Monteiro, A. R. (2015). *The teaching profession: Present and future* (pp. 47-60). Springer International Publishing.

Preparing to be a teacher

(<https://drive.google.com/file/d/1hEg15AS3dGtVxhYwGVrBtwQ7896qK0Fe/view?usp=sharing>)

## Vocational Pedagogical Course

### VPED 100 Principles and methods of technical instruction

**Course:** Principles and Methods of technical Instruction

**Code:** VPED 100

**Credit Hours:** 2 (1 TH + 1 PR)

**Nature:** Theory and Practical

#### Course Description

This course is designed to provide theoretical and practical exposure to students in planning teaching, learning, and evaluation by applying various methods based on the subject matter. It helps develop a sound understanding to get to know the learners in terms of their background, prior learning, sociocultural contexts, etc. Similarly, the course also centralizes its goal to help students locate and acquire the resources: dry and wet labs, studios, maker space, workshop, etc. Moreover, another primary focus is to help students be aware of impending difficulties, questions, disruptive behavior, conflicts, and ways of handling them by being mindful of different learning difficulties the learners face in skills development. Also, the course focuses on promoting independent learning among the learners through self-discovery, problem-solving, and product development.

#### Major Learning Outcomes

- Demonstrate a sound understanding of TVET overview and various technical instructions
- Apply various methods while planning for teaching and evaluation
- Explore multiple strategies to understand students based on their background, prior learning, sociocultural contexts
- Compare and contrast among various resources
- Explore ideas to handle the behaviors of students
- Develop skills in being mindful of different learning difficulties faced by learners and help them to progress
- Apply the principles of making students independent learners
- Develop lessons that use transversal skills for their vocational areas

#### Contents in detail with Specific Objectives

| Contents                                                                                  | Specific Objectives              | Resources                                                                                                                       |
|-------------------------------------------------------------------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Unit I: Overview of Technical and Vocational Education and Training (TVET)-3 hours</b> |                                  |                                                                                                                                 |
| • Fundamental concept of TVET                                                             | • Understand the concept of TVET | • <a href="https://unevoc.unesco.org/wt-db/worldtvtdatabase_npl_en">https://unevoc.unesco.org/wt-db/worldtvtdatabase_npl_en</a> |

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| <ul style="list-style-type: none"> <li>History and progress of TVET in Nepal</li> <li>Learning (formal, Non-formal, In-formal learning)</li> </ul>                                                              | <ul style="list-style-type: none"> <li>Investigate and research on a historical root and foundation of TVET in Nepal</li> <li>Compare the formal, non-formal, and in-formal TVET</li> </ul>                                                                                                                                             | <ul style="list-style-type: none"> <li><a href="#">.pdf#page=8&amp;zoom=100,116,160</a></li> <li><a href="https://unevoc.unesco.org/home/Dynamic+TVET+Country+Profiles/country=NPL">https://unevoc.unesco.org/home/Dynamic+TVET+Country+Profiles/country=NPL</a></li> </ul>                                                                         |
| <b>Unit II: Introduction to Technical Instruction -5 hours</b>                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                     |
| <ul style="list-style-type: none"> <li>Methods and orientations,</li> <li>Experiential learning of John Dewey</li> <li>Developing competencies and outcomes</li> <li>Developing tasks and activities</li> </ul> | <ul style="list-style-type: none"> <li>Conceptualize the method and orientation of Technical Instruction.</li> <li>Explain and use the experiential learning of John Dewey.</li> <li>Explain and develop the competencies and outcomes for ICT Instruction</li> <li>Develop the tasks and activities for the ICT instruction</li> </ul> | <ul style="list-style-type: none"> <li><a href="https://eacea.ec.europa.eu/national-policies/eurydice/content/teaching-and-learning-vocational-and-technical-upper-secondary-education_en">https://eacea.ec.europa.eu/national-policies/eurydice/content/teaching-and-learning-vocational-and-technical-upper-secondary-education_en</a></li> </ul> |
| <b>Unit III: Understanding Students in Educational Instruction -5 hours</b>                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                     |
| <ul style="list-style-type: none"> <li>Developing students' profiles</li> <li>Assessing prior knowledge</li> <li>Planning for remedial instruction</li> <li>Addressing diversities as asset</li> </ul>          | <ul style="list-style-type: none"> <li>Explain the student's profiles</li> <li>Develop the student profile</li> <li>Examine the prior knowledge of the learner</li> <li>Apply for the remedial instruction in classroom</li> <li>Apply the techniques for the diversities of the learner as asset in the classroom.</li> </ul>          | <ul style="list-style-type: none"> <li><a href="https://www.nap.edu/read/5287/chapter/9">https://www.nap.edu/read/5287/chapter/9</a></li> </ul>                                                                                                                                                                                                     |
| <b>Unit IV: Resource Materials for the Instructional Activities -4 hours</b>                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                     |

|                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                   |
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| <ul style="list-style-type: none"> <li>• Projected and non-projected materials for technical instruction</li> <li>• Instructional media</li> <li>• Apps and Learning Management System</li> </ul>        | <ul style="list-style-type: none"> <li>• Define the projected and non-projected material for the technical instruction</li> <li>• Explain and use of instructional media in instructional activities.</li> <li>• Experiment of apps and learning management system (Google class, Moodle)</li> </ul> | <ul style="list-style-type: none"> <li>• <a href="https://teaching.unl.edu/course-design/flexible-hybrid/instructional-materials/">https://teaching.unl.edu/course-design/flexible-hybrid/instructional-materials/</a></li> </ul> |
| <b>Unit V: Classroom Management-5 hours</b>                                                                                                                                                              |                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                   |
| <ul style="list-style-type: none"> <li>• Management for efficiency and management for equity</li> <li>• Promoting participation</li> <li>• Ensuring the success for all - reaching out to all</li> </ul> | <ul style="list-style-type: none"> <li>• Classroom management strategies</li> <li>• Define about management for efficiency and management for equity</li> <li>• Explain the steps for promoting participation</li> <li>• Explain and examine the success for all -reaching out to all</li> </ul>     | <ul style="list-style-type: none"> <li>• <a href="https://web.calstatela.edu/faculty/jshindl/cm/Chapter11pedagogy-final.htm">https://web.calstatela.edu/faculty/jshindl/cm/Chapter11pedagogy-final.htm</a></li> </ul>             |
| <b>Unit VI: Work and learning-4 hours</b>                                                                                                                                                                |                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                   |
| <ul style="list-style-type: none"> <li>• Situated Learning: Work as context for learning</li> <li>• Complexity of work and learning</li> <li>• Different forms of work-based learning</li> </ul>         | <ul style="list-style-type: none"> <li>• Define about situated learning: work as context for learning</li> <li>• Understand the different form of work-based learning</li> <li>• Critically examine the complexity of work and learning</li> </ul>                                                   | <a href="https://doi.org/10.1108/13665621311316447">https://doi.org/10.1108/13665621311316447</a>                                                                                                                                 |

|                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                           |
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| <ul style="list-style-type: none"> <li>Developing work-based learning modules</li> </ul>                                                                                                                                               | <ul style="list-style-type: none"> <li>Explain and compare about different forms of work-based learning</li> <li>Develop the work-based learning module.</li> </ul>                                                                                                                                                               |                                                                                                                                                                                                                                           |
| <b>Unit VII: Transversal Skills for technical and vocational education-4 hours</b>                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                           |
| <ul style="list-style-type: none"> <li>The 4 C framework—Communication, Critical Thinking, Creativity and Collaboration</li> <li>The blend of soft and hard skills</li> <li>Integrating transversal skills for TVET lessons</li> </ul> | <ul style="list-style-type: none"> <li>Define and critically examine 4 C framework-communication, critical thinking, creativity, and collaboration</li> <li>Apply the blending of soft and hard skills</li> <li>Evaluate the different roles and responsibilities for integrating transversal skill for TVET teaching.</li> </ul> | <ul style="list-style-type: none"> <li><a href="https://bangkok.unesco.org/content/transversal-skills-tvet-pedagogies-and-assessment">https://bangkok.unesco.org/content/transversal-skills-tvet-pedagogies-and-assessment</a></li> </ul> |

### **Methodology and Teaching Techniques**

The facilitator(s) can use multiple methods or strategies to be teaching and learning listed below. Students must be actively engaged in each and every learning activity. The frequent sharing session and reflection activities should be conducted. So, teaching and learning should be based on constructivist model.

- Lecture Method
- Inquiry-based approach
- Inductive & Deductive Methods
- Demonstration approach
- ICT integrated approach
- Collaborative learning (Peer and group learning)
- Discovery and investigation approach
- Project-based learning

### **Major Teaching and Learning Materials**

Materials are essential to communicate knowledge and skills among learners. Rich learning resources and materials must be used during teaching and learning activities. The

facilitator(s) is/are suggested to use the following resource materials based upon the nature of the course contents. Also, the facilitator is responsible for developing these learning resources.

- Teacher's Notes and Hands-out
- PPT developed by the facilitator
- Daily used materials
- Chart and cardboard papers
- Laptop/Desktop and projector
- Books or printed materials
- Other materials

### **Evaluation Scheme**

**In-semester Assessment:** 50% (Home Assignments and classwork). While doing this assessment, the facilitator(s) is/are recommended to use more than one strategy listed below.

- Attendance (Compulsory, 70%)
- Home Assignments (which may include the following)
- Reflective writings (These tasks encourage students to explore and reflect on the past experiences/current practices related to teaching profession, develop possible strategies and alternatives to improve the practices, be imaginative to portray their understanding/creativity that stimulate students for bringing their vision/s as agents of change)
- Presentations and demonstration (Individual and group presentations on topics related to the contents)
- Regular Practical Tasks (Teaching and Learning simulations)
- Forum Discussion/Collaborative group work
- Written test
- Quizzes
- Project work (Field visit related to teaching profession followed by report writing)
- Viva-voce
- Open book test

**Note:** During every formative assessment task, a continuous and constructive feedback must be provided to improve the learning of students. Students must be given chances for improving their grades and learning. So, a Continuous Assessment System (CAS) should be followed.

**End-semester Assessment:** 50% (Examination conducted by Kathmandu University)

### **Resources**

- Rus, R. C., Husain, M. A. M., Hanapi, Z., & Mamat, A. B. (2020). TVETagogy: Teaching and Facilitating Framework (PDPC) for Technical and Vocational Education and Training (TVET). *International Journal Of Academic Research In Business And Social Sciences*, 10(3).
- Pavlova, M., & Chen, C. S. (2019). Facilitating the development of students' generic green skills in TVET: an ESD pedagogical model. *TVET@ Asia*, 12, 1-21.
- Maclean, R., & Wilson, D. (2009). *International handbook of education for the changing world of work: Bridging academic and vocational learning* (Vol. 1). C. A. Chinien (Ed.). Dordrecht: Springer Netherlands

## Occupational/Specialization Course

### ITEX 100 Information System

**Course:** Information System

**Code:** ITEX 100

**Credit Hours:** 3 (2 TH + 1 PR)

**Nature:** Theory and Practical

#### Course Description and Goals

The course helps learners develop fundamental knowledge and understanding relating to data management, Network technology, Computer security system, emerging trends and consider the impact of emerging trends in information technology with case studies.

#### Learning Outcomes

On completion of this course, the students will be able to:

- Understanding historical background of computing
- Understand the computer systems in an organization
- Familiarity of computer software
- Fundamental understanding of network technology
- Security of computer-based information
- Become conversant with system design issues
- Demonstrate an awareness of Human Computer Interaction and the use of multimedia and hypermedia
- Develop awareness of the basic ideas behind using a computer to store and manipulate data
- Display knowledge of data analysis and modelling techniques
- Discuss various database management architectures
- Understand the need for quality assurance/security in computer systems development and operation

#### Content with Specific Objectives

| Specific objectives                                                                                                                                   | Contents                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Recall the historical background of computing</li><li>• To understand trend of computing technology</li></ul> | <b>Unit One: Historical background of computer (2 hrs.)</b> <ul style="list-style-type: none"><li>• Evolution of computing technology</li><li>• Technological aspect of generation of computer</li></ul> |
| <ul style="list-style-type: none"><li>• Understand fundamental working units of computer with peripherals devices and memory units.</li></ul>         | <b>Unit Two: Basic computer organization and input output devices (8 hrs.)</b> <ul style="list-style-type: none"><li>• Computer architecture</li><li>• Components of Computer System,</li></ul>          |

|                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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|                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>• Central Processing Unit (CPU)</li> <li>• Common Input and output devices</li> <li>• Types of Computer Memory</li> <li>• Concepts of Hardware and Software</li> <li>• Video Display Unit (CRT, LCD, LED, DVST, QLED, OLED)</li> <li>• Types of Printers and Scanners, working principles</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <ul style="list-style-type: none"> <li>• Understand types of software</li> <li>• Compare different types of software used in computer and their application.</li> </ul>                                                                              | <b>Unit Three: Computer software (4 hrs.)</b> <ul style="list-style-type: none"> <li>• System Software</li> <li>• Operating System</li> <li>• Language Processor</li> <li>• Device Drivers</li> <li>• Application Software</li> <li>• Utility Software</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <ul style="list-style-type: none"> <li>• Illustrate internet with associated devices and their arrangements.</li> <li>• Demonstrate with web browsing technology</li> <li>• Understand computer in network</li> <li>• Identify google map</li> </ul> | <b>Unit Four: Computer network (8 hrs.)</b> <ul style="list-style-type: none"> <li>• Transmission modes</li> <li>• Communication media</li> <li>• Concept of Internet and its services</li> <li>• Requirements for connection to internet</li> <li>• Connecting to internet; ISP;</li> <li>• Network connectivity devices</li> <li>• Basics of internet connectivity related troubleshooting, World Wide Web.</li> <li>• Web Browsing software's, Search Engines.</li> <li>• Understanding URL; Domain name; IP Address.</li> <li>• Use of e-governance website</li> <li>• Network architecture</li> <li>• Basic of Computer networks; LAN, WAN.</li> <li>• Topologies, Advantages, and disadvantages.</li> <li>• Google map familiarity of tools of GIS</li> </ul> |
| <ul style="list-style-type: none"> <li>• Build data security in network</li> <li>• Identify types of threats with issues, control policies</li> </ul>                                                                                                | <b>Unit Five: Computer security and information privacy (7 hrs.)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

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| <ul style="list-style-type: none"> <li>• Understand types of access control</li> <li>• Understand of malicious software</li> <li>• Plan and recover for data</li> </ul> | <ul style="list-style-type: none"> <li>• Introduction to computer security</li> <li>• Basic components of security (Confidentiality, Integrity and Availability), Security threats (Snooping, Modification, Masquerading, repudiation of origin, denial of receipt, Delay, Denial of service)</li> <li>• Issues with security (Operational issues, human issues).</li> <li>• Security Policies, Type of security policy</li> <li>• Access control</li> <li>• Type of access control (Introduction to MAC, DAC, Originator Controlled Access Control, Role Based Access Control)</li> <li>• Overview of the Bell-LaPadula Model and Biba integrity model.</li> <li>• System vulnerability and abuse: vulnerability; malicious software;</li> <li>• Hackers and computer crime; Internal Threats: employees.</li> <li>• Value of security and control; legal and regulatory requirements for electronic records management; electronic evidence and computer forensics</li> <li>• Establishing a framework for security and control; information systems controls; risk assessment; security policy.</li> <li>• Disaster recovery planning and business continuity planning.</li> <li>• The role of auditing Hand-n MIS projects</li> </ul> |
| <ul style="list-style-type: none"> <li>• Familiarize with emerging technology in computing</li> </ul>                                                                   | <p><b>Unit Six: Emerging technology (5 hrs.)</b></p> <ul style="list-style-type: none"> <li>• Machine learning</li> <li>• Quantum computing</li> <li>• Metaverse</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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| <ul style="list-style-type: none"> <li>• Develop computing technology in business and governance</li> <li>• Understand multimedia and application</li> </ul>    | <b>Unit Seven: E-Commerce / Digital Services (5 hrs.)</b> <ul style="list-style-type: none"> <li>• E-commerce</li> <li>• Trade cycles</li> <li>• Modes of payment</li> <li>• E-Governance: G2C, G2B, G2E, G2E</li> <li>• E governance practice in Nepal</li> <li>• Multimedia and applications</li> </ul>                                                                                                                                                                                                                                                                                                                                    |
| <ul style="list-style-type: none"> <li>• Build concept of database</li> <li>• Apply DBMS in Business intelligence</li> <li>• Understand MIS projects</li> </ul> | <b>Unit Eight: Databases and Information Management: (4 hrs.)</b> <ul style="list-style-type: none"> <li>• Organizing Data in a traditional file environment; file organization concepts; problems with the traditional file environment</li> <li>• The database approach to data management: Database Management Systems; capabilities of database Management systems; designing database</li> <li>• Using databases to improve business performance and decision making; data warehouses; tools for business intelligence: Multidimensional data analysis and data mining</li> <li>• Interactive session: Hands-on MIS projects</li> </ul> |
| <ul style="list-style-type: none"> <li>• Understand and analyze of computing technology</li> </ul>                                                              | <b>Unit Nine: Case studies based on Application of Information system (2 hrs.)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

### Major Teaching and Learning Strategy

The practical exercise shall include following types of projects on designing of information system.

- Lecture and Practical
- Group Discussion
- Presentation by Students and Teachers
- Lab work in computer Lab
- Project work

### Assessment Plan

- a. In-semester :50%
- b. End-semester:50%

## References

- Information Systems Today Managing in the Digital World Joe Valacich Washington State University Christoph Schneider City University of Hong Kong
- Managing With Information System, J. Kanter, PHI, Latest edition
- Introduction to Search Engine Optimization A Guide for Absolute Beginners — Todd Kelsey with Brandon Lyon
- Computer Fundamentals: Architecture and Organization
- Cybersecurity Essentials Charles J. Brooks, Christopher Grow, Philip Craig, Donald Short
- The essentials of computer organization and architecture/Linda Null and Julia Lobur

## ITEX 101 Software Development

**Course:** Software Development

**Course Code:** ITEX 101

**Credit Hours:** 3 (2 TH + 1 PR)

**Nature:** Theory and Practical

### Course Description

- Learners will understand fundamental concepts of the programming process, consider issues related to various phases of software development, be introduced to different types of programming concepts, able to understand Linux programming, develop concepts of software maintenance and reuse, basic concepts of big data with detailed software practices.

### Learning outcome

- Develop the concept of software and basic digital rights management, software licensing and certification related to software development
- Exhibit the skills of tools and techniques of software development
- Demonstrate the basic Linux programming techniques
- Apply the concept of big data with detailed case study

| Specific objective                                                                                                                                                                                                                                      | Contents                                                                                                                                                                                                                                                                                                                                                                                              | Contact hours |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <ul style="list-style-type: none"><li>• To develop the concept, history evolution and types of software</li><li>• To describe the basic software patents, digital rights management, software licensing and software – engineer certification</li></ul> | <b>1. Introduction to software</b><br>1.1 Concept of software:<br>1.2 History of software<br>1.3 Evolution of software<br>1.4 Types of software<br>1.5 Software patents<br>1.6 Digital rights management<br>1.7 Software licensing<br>1.8 Software-engineer certification                                                                                                                             | 7             |
| <ul style="list-style-type: none"><li>• To describe the techniques of software development</li><li>• To explain the student about the basic programming language used in software development</li></ul>                                                 | <b>2. Techniques of software development</b><br>2.1 Software development process<br>2.1.1 Agile development methodology<br>2.1.2 DevOps deployment methodology<br>2.1.3 Waterfall development method<br>2.1.4 Rapid application development<br>2.1.5 Modular programming techniques<br>2.1.6 Evolutionary and prototyping model<br>2.2 Programming language used in software development (case study) | 7             |
| <ul style="list-style-type: none"><li>• To demonstrate the basic file process and utilities</li><li>• To illustrate the various Linux programming techniques</li></ul>                                                                                  | <b>3. Files process and shells</b><br>3.1 Command-line utilities<br>3.2 File editing<br>3.3 Shell scripting<br>3.4 String processing<br>3.5 Regular expressions<br>Web scripting (CGI, http, html)                                                                                                                                                                                                    | 6             |

|                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                      |   |
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| <ul style="list-style-type: none"> <li>To identify the various programming tools associated with software development like standard API, IDE, debuggers, profilers, linkers, libraries, compilation managers and version control systems</li> </ul> | <b>4. Programming tools</b><br>4.1 Standard API<br>4.2 Integrated development environment<br>4.3 Debuggers<br>4.4 Profilers<br>4.5 Linkers<br>4.6 Libraries<br>4.7 Compilation managers<br>Version-control systems                                                                                                                   | 8 |
| <ul style="list-style-type: none"> <li>To interpret the maintenance and reuse of software</li> </ul>                                                                                                                                                | <b>5. Maintenance and Reuse</b><br>5.1 Need for maintenance<br>5.2 Challenges of software maintenance<br>5.3 Categories of software maintenance<br>5.4 Introduction to software reuse<br>5.5 Re-use based software engineering<br>5.6 Benefits of software reuse<br>5.7 Challenges of software reuse<br>Techniques of software reuse | 7 |
| <ul style="list-style-type: none"> <li>To describe the concept of big data</li> </ul>                                                                                                                                                               | <b>6. Big data</b><br>6.1 Introduction<br>6.2 Characteristic of big data<br>6.3 Storing, selecting and processing big data<br>6.4 Need of big data<br>6.5 Tools of big data<br>6.6 Application of big data<br>6.7 Risk of big data<br>6.8 Future of big data<br>Case studies on big data                                             | 8 |
| <ul style="list-style-type: none"> <li>To apply the detailed study based on current software trends</li> </ul>                                                                                                                                      | Case studies                                                                                                                                                                                                                                                                                                                         | 2 |

### Major Teaching and Learning Strategy

- Lecture and Discussion
- Presentation and Demonstration
- Practical Work with Computer and in Computer Lab
- Project Work

### Assessment Plan

- c. In-semester :50%
- d. End-semester:50%

## References

- Effective software maintenance and evolution: A reuse-based approach by Stanislaw Jarzabek.
- Linux For Beginners: The Ultimate Guide to The Linux Operating System & Linux by Vardy Adam.
- Software Development from A to Z: A Deep Dive into All the Roles Involved in the Creation of Software Olga Filipova & Rui Vilão.
- Big Data in Practice: How 45 Successful Companies Used Big Data Analytics to Deliver by Bernard Marr.

## ITEX 102 Computer Programming Using C

**Course Title:** Computer Programming using C

**Code:** ITEX 102

**Credit Hours:** 3 (2 TH + 1 PR)

**Nature:** Theory +Practical

### Course Description and Goals

This course introduces C programming concepts and software development using C Programming Language.

### Learning Outcomes

- Design, implement, and document (appropriately) efficient algorithms for a C application using a structured approach to programming.
- Explain and apply the concept of a procedural programming paradigm and apply the associated design principles to a range of problem solutions.
- Illustrate the use of C as both an application and a systems programming language by being able to design, implement, and document efficient algorithms for a C application.
- Explain the structure, correct use of and implementation of appropriate advanced data structures for a range of scenarios.

### Content with Specific Objectives

| Specific objectives                                                                                                                                              | Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <ul style="list-style-type: none"><li>• Understanding and drawing concept of stepwise representation with pictorial representation of computer program</li></ul> | <b>Unit/Module One: Algorithm Design with Pseudocode and Graphical Representation (4 Hrs.)</b> <ul style="list-style-type: none"><li>○ Algorithms<ul style="list-style-type: none"><li>○ Expressing Algorithms</li><li>○ Benefits of Using Algorithms</li><li>○ General Approaches in Algorithm Design</li><li>○ Analysis of Algorithms</li></ul></li><li>○ Flowcharts<ul style="list-style-type: none"><li>○ Advantages of Using Flowcharts</li><li>○ Limitations of Using Flowcharts</li><li>○ When to Use Flowcharts</li><li>○ Flowchart Symbols &amp; Guidelines</li><li>○ Types of Flowcharts</li></ul></li><li>○ Steps in Solving a Problem</li><li>○ Activities involved in Program Design</li></ul> |
| <ul style="list-style-type: none"><li>• Outline concepts of programming</li></ul>                                                                                | <b>Unit Two: Introduction (8 Hrs.)</b> <ul style="list-style-type: none"><li>○ Introduction to C Programming<ul style="list-style-type: none"><li>• Introduction to programming language</li><li>• Application aspects</li><li>• Lexical elements, data types and operators</li><li>• Programming structure and construction.<ul style="list-style-type: none"><li>○ Standard library and function.</li><li>○ Programming Style.</li><li>○ Program structure.</li></ul></li><li>• Writing and Running a C program.</li></ul></li></ul>                                                                                                                                                                      |
| <ul style="list-style-type: none"><li>• Develop logical and iteration practice of computer program</li></ul>                                                     | <b>Unit/Module Three – Programming Construction (8 Hrs.)</b> <ul style="list-style-type: none"><li>• Programming construction - Conditional Controls</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

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|                                                                                                                                                         | <ul style="list-style-type: none"> <li>○ IF statement</li> <li>○ IF – Else Statement</li> <li>○ Nested IF – Else</li> <li>○ Switch case statements</li> <li>• Programming construction - Iterative Controls <ul style="list-style-type: none"> <li>○ While loop</li> <li>○ For loop</li> <li>○ Do-while loop</li> <li>○ Nested loops</li> </ul> </li> </ul>                                                                                               |
| <ul style="list-style-type: none"> <li>• Demonstrate similar data handling with address.</li> <li>• Demonstrate concept of data referencing</li> </ul>  | <b>Unit/Module Four: Arrays &amp; Pointers (7 Hrs.)</b> <ul style="list-style-type: none"> <li>• Arrays <ul style="list-style-type: none"> <li>○ Introduction to arrays</li> <li>○ One Dimensional</li> <li>○ Two Dimensional</li> </ul> </li> <li>• Pointers <ul style="list-style-type: none"> <li>○ Introduction to pointers</li> <li>○ Declaring and accessing data through pointers.</li> </ul> </li> </ul>                                          |
| <ul style="list-style-type: none"> <li>• Understand and apply modular programming</li> <li>• Illustrate storage class of data</li> </ul>                | <b>Unit/Module Five: Modular Programming in C(Functions) (7 Hrs.)</b> <ul style="list-style-type: none"> <li>• Concepts of Functions</li> <li>• Call by value and Call by reference <ul style="list-style-type: none"> <li>○ Recursion concept</li> <li>○ Passing arrays on functions</li> </ul> </li> <li>• Storage classes <ul style="list-style-type: none"> <li>○ Auto</li> <li>○ Extern</li> <li>○ Register</li> <li>○ Static</li> </ul> </li> </ul> |
| <ul style="list-style-type: none"> <li>• Build structure and memory allocation concept</li> <li>• Apply dynamic memory for memory management</li> </ul> | <b>Unit/Module Six: Data Structures (7 Hrs.)</b> <ul style="list-style-type: none"> <li>• Struct</li> <li>• Union</li> <li>• Bitfields</li> <li>• Dynamic Memory Allocation</li> <li>• Linked List</li> </ul>                                                                                                                                                                                                                                             |
| <ul style="list-style-type: none"> <li>• Outline concepts of permanent storage of data with addition and updating record files</li> </ul>               | <b>Unit/Module Seven: Record &amp; Files (4 Hrs.)</b> <ul style="list-style-type: none"> <li>• Text and sequential files</li> <li>• Creating and accessing files</li> <li>• Validation, update and file maintenance algorithms</li> </ul>                                                                                                                                                                                                                 |

#### Major Teaching and Learning Strategy

- Example methods and simple projects
- At least 11 Lab works.
- Lectures and Presentation
- Group Discussions
- Presentation by Students
- Inquiry-based Learning

**Assessment Plan**

- a. In-semester: 50%**
- b. End-semester: 50%**

**References**

- Balagurusamy, E. (2019). *Programming in ANSI C* (8th ed.). McGraw Hill Education.
- Paul, D., & Harvey, D. (2022). *C: How to Program* (9<sup>th</sup> Ed.). Pearson Education.
- Thareja, R. (2018). *Data Structures Using C* (2<sup>nd</sup> ed.). Oxford University Press.

## ITEX 113 Mathematical Concepts for Computing

Course Title: Mathematical Concepts for Computing

Course Code: ITEX 113

Credit Hours: 3 (2 Th + 1 PR) Nature: Theory + Practical

### Course Description and Goals

This course provides key mathematical concepts required in computing. Topics include number base systems, set theory, relations and functions, linear algebra, vectors and vector spaces, and counting techniques. Emphasis is placed on the application of mathematics in programming, data representation, graphics modeling, and security. Hands-on experience with computational tools such as Python and Excel helps students connect mathematical concepts to real-world IT applications.

### Learning Outcomes

- Represent data in various number systems and perform binary arithmetic operations.
- Apply set operations, relations, and functions in computing contexts.
- Use matrices and linear algebra concepts for system modeling and computer graphics.
- Analyze vectors and vector spaces including basis, dimension, and eigenvalues.
- Apply permutations and combinations for problem-solving in IT security and algorithms.
- Use Python, Excel, and visualization tools for mathematical computation and graphical interpretation.
- Communicate mathematical ideas clearly in written and visual form.

### Content with Specific Objectives

| Specific Objectives                                                                                                                                      | Contents                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Convert numbers between bases</li><li>• Perform binary arithmetic</li><li>• Explain storage of numbers</li></ul> | <b>Unit One: Number Base System (6 hours)</b> <ul style="list-style-type: none"><li>○ Base systems and conversions (2,8,10,16)</li><li>○ Binary arithmetic</li><li>○ Complements</li><li>○ IT Application: Low-level data representation</li></ul> |
| <ul style="list-style-type: none"><li>• Use set operations</li></ul>                                                                                     | <b>Unit Two: Set Theory (7 hours)</b>                                                                                                                                                                                                              |

|                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <ul style="list-style-type: none"> <li>• Apply Venn diagrams</li> <li>• Apply DBMS set analogy</li> </ul>                                              | <ul style="list-style-type: none"> <li>○ Sets, operations, Venn diagrams (three sets)</li> <li>○ Laws of sets</li> <li>○ IT Applications: DBMS set operators</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <ul style="list-style-type: none"> <li>• Classify relations</li> <li>• Apply elementary functions</li> <li>• Connect functions to computing</li> </ul> | <p><b>Unit Three: Relations &amp; Functions (10 hours)</b></p> <p>Relations</p> <ul style="list-style-type: none"> <li>○ Introduction</li> <li>○ Properties</li> <li>○ Equivalence relation</li> </ul> <p>Functions</p> <ul style="list-style-type: none"> <li>○ Introduction and types</li> <li>○ Inverse of function and compositions</li> </ul> <p>Elementary functions:</p> <ul style="list-style-type: none"> <li>○ linear, quadratic, exponential, logarithmic and trigonometric with example and problems</li> <li>○ graphical interpretation of functions</li> </ul> <p>IT Applications: Key mapping and relational modeling</p> |
| <ul style="list-style-type: none"> <li>• Perform matrix operations</li> <li>• Solve linear equations</li> </ul>                                        | <p><b>Unit Four: Linear Algebra (9 hours)</b></p> <p>Matrices</p> <ul style="list-style-type: none"> <li>○ operations: addition, subtraction, multiplication</li> </ul> <p>Determinants and Inverse</p> <ul style="list-style-type: none"> <li>○ Determinant evaluation</li> <li>○ Properties of determinant</li> <li>○ Inverse of matrix</li> </ul>                                                                                                                                                                                                                                                                                     |

|                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                              | <p>System of linear equation</p> <ul style="list-style-type: none"> <li>○ Gauss elimination (three variables)</li> <li>○ Consistency and rank</li> <li>○ Graphs</li> </ul> <p>IT Applications: Graphics transformation, encoding</p>                                                                                                                                                                                                                                                                                                                                                                                                            |
| <ul style="list-style-type: none"> <li>• Interpret vectors in 2D/3D</li> <li>• Understand vector space properties</li> <li>• Apply eigen concepts</li> </ul> | <p><b>Unit Five: Vectors &amp; Vector Space (10 hours)</b></p> <p>Vector in 2D and 3D</p> <ul style="list-style-type: none"> <li>○ Vector operations</li> <li>○ dot product</li> <li>○ cross product and geometric interpretation</li> </ul> <p>Vector space concept</p> <ul style="list-style-type: none"> <li>○ Definition and examples</li> <li>○ subspace,</li> </ul> <p>Basis and Dimension</p> <ul style="list-style-type: none"> <li>○ Linear combination of vector</li> <li>○ Linear dependence and independence</li> <li>○ Basis and dimension</li> </ul> <p>Eigenvalue and eigenvector</p> <p>IT Applications: 2D and 3D modeling</p> |
| <ul style="list-style-type: none"> <li>• Apply counting methods</li> <li>• Solve IT arrangement problems</li> </ul>                                          | <p><b>Unit Six: Permutations &amp; Combinations (6 hours)</b></p> <p>Counting principle</p> <ul style="list-style-type: none"> <li>○ Factorial and arrangements</li> <li>○ Permutations,</li> </ul> <p>Combinations</p> <p>IT Applications: Cryptography, data organization</p>                                                                                                                                                                                                                                                                                                                                                                 |

## Major Teaching and Learning Strategy

### 1. Interactive Lectures

- Short theory presentations with emphasis on how each mathematical concept supports computing technologies
- Frequent “think-pair-share” prompts to encourage analytical participation
- Use of visual representations (graphs, vector illustrations, matrices, transformations)

### 2. Demonstration and Guided Practice

- Instructor-led stepwise problem-solving on the board and screen
- Demonstrations using Python, Excel, Desmos, or GeoGebra
- Students replicate demonstration tasks immediately after concepts are introduced

### 3. Collaborative Learning

- Group worksheets and mini-projects to promote peer learning
- Students compare strategies, justify choices, and explain results
- Team-based practical sessions to simulate real collaborative IT work

### 4. Case-Based and Contextual Learning

- Real IT application scenarios such as:
  - Binary storage and hardware data representation
  - Graphics transformation matrices
  - Eigen decomposition in animation and modeling
  - Counting principles in cybersecurity
- Helps students connect mathematical concepts with professional computing tasks

### 5. Hands-on Computational Learning

- Minimum 16 hours of practical sessions using:
  - Python with NumPy/Matplotlib
  - Excel and visualization tools

- Direct reinforcement of theory through computational exploration

## **6. Inquiry and Problem-Solving Approach**

- Students identify problem requirements, evaluate methods, and justify solutions
- Analytical tasks tied to algorithmic thinking to strengthen logic skills
- Step-by-step progression from structured to open-ended tasks

## **7. Assessment for Learning**

- Regular quizzes, concept check activities, and instant feedback tools
- Weekly lab assessments measuring accuracy + reasoning
- Formative assessment to guide progression toward final project

## **8. Self and Peer Evaluation**

- Students review peers' lab outputs and project components
- Enhances critical evaluation and communication skills
- Encourages responsibility for accuracy and clarity in computation

## **9. Use of Digital Learning Resources**

- LMS materials, recorded demonstrations, online documentation
- Academic articles and IT examples to strengthen application perspective
- Platforms such as GitHub for version tracking and collaboration in assignments

## **Practical and Project Work**

### **Computational Labs (16 Hours Minimum)**

Students will complete the following practical activities:

#### **1. Lab 1: Number System and Binary Arithmetic (2 Hours)**

- Python program for base conversions (2,8,10,16)
- Binary addition & subtraction
- 1's and 2's complement subtraction
- Verification using built-in functions

#### **2. Lab 2: Set Operations using Python/Excel (2 Hours)**

- Union, intersection, difference, symmetric difference
- De Morgan's laws validation

- Venn diagrams visualization
- Database segmentation case task

### 3. Lab 3: Relations & Functions Visualization (3 Hours)

- Matrix representation of relations
- Check: reflexive, symmetric, transitive
- Plot linear, quadratic, exponential, logarithmic functions in Geogebra / Matlab
- Graph transformations

### 4. Lab 4: Matrix Operations & System Solving (3 Hours)

- Matrix arithmetic using NumPy/Matlab
- Determinant & inverse calculation
- Solving systems with Gaussian elimination
- Simple networking application

### 5. Lab 5: Vector Algebra & Cross Product (3 Hours)

- Dot and cross product computation
- Linear combination & span
- Rank-based independence test
- Vector visualization

### 6. Lab 6: IT Applications Mini Project (3 Hours)

- Project integrating mathematical concepts into computing
- Examples: password security, 3D transformations, relational mapping, cryptographic modeling
- Project presentation and report

### Laboratory Assessment Rubric

- Program Accuracy (40%): Correct computation and logic implementation
- Documentation & Reporting (25%): Explanation, screenshots, clarity of code
- Visualization & Interpretation (20%): Correct outputs and interpretation
- Presentation & Collaboration (15%): Professional communication and teamwork

### Assessment Plan

Total Assessment Weight: In-semester 50% | End-semester 50%

| Component          | Marks | Description                      |
|--------------------|-------|----------------------------------|
| Assignments        | 15    | Weekly tasks and problem sets    |
| Quizzes            | 10    | Short tests on core topics       |
| Lab Work & Reports | 15    | Computation-based practical work |

|                        |    |                                 |
|------------------------|----|---------------------------------|
| Project & Presentation | 10 | Mini-project with demonstration |
|------------------------|----|---------------------------------|

### End-Semester Examination Structure:

- MCQ: 10 Questions  $\times$  1 mark = 10
- SAQ: 6 Questions  $\times$  4 marks = 24 (out of 7)
- LAQ: 2 Questions  $\times$  8 marks = 16 (out of 3)

### Learning Units Mapping to Assessment

| Learning Units | Theory hrs | Practical hrs | MCQs | SAQs | LAQs |
|----------------|------------|---------------|------|------|------|
| Unit One       | 4          | 2             | 1    | 1    | -    |
| Unit Two       | 5          | 3             | 2    | 1    | -    |
| Unit Three     | 7          | 3             | 2    | 1    | 1    |
| Unit Four      | 6          | 3             | 2    | 1    | 1    |
| Unit Five      | 7          | 3             | 2    | 2    | 1    |
| Unit Six       | 4          | 2             | 1    | 1    | -    |
| Total          | 32         | 16            | 10   | 7    | 3    |

### References

Johnsonbaugh, R. (2018). *Discrete Mathematics* (8th ed.). Pearson.

Lay, D. C. (2015). *Linear Algebra and Its Applications* (5th ed.). Pearson.

Strang, G. (2016). *Introduction to Linear Algebra* (5th ed.). Wellesley-Cambridge Press.

Epp, S. S. (2020). *Discrete Mathematics with Applications* (4th ed.). Cengage Learning.

Jenkyns, T., & Stephenson, B. (2018). *Fundamentals of discrete math for computer science: A problem-solving primer* (2<sup>nd</sup> ed.). Springer.

## ITEX 105 Professional Issues in IS Practices

**Course Title:** Professional Issues in IS Practices

**Course Code:** ITEX 105

**Credit Hours:** 2 (1 TH + 1 PR)

### Course Description and Goals

This course is designed to provide knowledge and better understanding to our To-be IT Professionals regarding business, organization, management, accounting, IT laws and policy hence encouraging them to be more able to work efficiently in this unpredictable environment. Also, it aims to produce well rounded individuals who make a strong contribution to the workplace and show sensitivity to business needs, as it is vital in the modern workplace for Learners to understand more than the software alone.

### Learning Outcomes

On completion of this course, the students will be able to:

- know about the basics of Management, its theories, functions, and practices in modern workplaces
- understand the basic financial accounting related terms
- understand the IT policy of Nepal and laws related to internet

### Content with specific objectives

| Specific Objectives                                                                                                                                                                                                                                  | Contents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ol style="list-style-type: none"><li>2. To define organization, its goals</li><li>3. To describe Management, its challenges, theories and globalization</li><li>4. To give a clear understanding regarding various theories of Management</li></ol> | <p><b>Unit One – Introduction to Management (8 Lecture Hours)</b></p> <ol style="list-style-type: none"><li>1. Concept of Organization, Goal and goal displacement</li><li>2. Introduction to Management, its features &amp; importance<ol style="list-style-type: none"><li>2.1 Organization and Management</li><li>2.2 Primary functions of Management</li><li>2.3 Levels/Hierarchy of Management</li><li>2.4 Emerging challenges of Management</li><li>2.5 Managing in the Global Arena: Concept of Globalization and its effects</li></ol></li><li>3. Overview of Management<br/>Theories/Perspectives in Management<ol style="list-style-type: none"><li>3.1 Theories in Classical approach<ul style="list-style-type: none"><li>• Scientific Management Theory</li><li>• Administrative Management Theory</li></ul></li><li>3.2 Theories in Behavioral approach<ul style="list-style-type: none"><li>• Human Relations Theory</li><li>• Theory X and Theory Y</li></ul></li><li>3.3 Theories in Modern approach<ul style="list-style-type: none"><li>• System Management Theory</li><li>• Contingency Management Theory</li></ul></li></ol></li></ol> |

|                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <ol style="list-style-type: none"> <li>1. To develop the concept of managers, their roles, skills and effectiveness</li> <li>2. To identify motivation and motivating factors in the organization</li> <li>3. To understand leadership possibilities &amp; also to have awareness of his/her personal leadership style</li> </ol> | <p><b>Unit Two: Managers in the workplace (10 Lecture Hours)</b></p> <ol style="list-style-type: none"> <li>1. Concept of Managers and their workplace             <ol style="list-style-type: none"> <li>1.1 Functions of Managers</li> <li>1.2 Roles and skills of Managers</li> <li>1.3 Mintzberg's Managerial roles</li> <li>1.4 Efficiency and Effectiveness in Management</li> </ol> </li> <li>2. Planning:             <ol style="list-style-type: none"> <li>2.1 Meaning &amp; levels of planning,</li> <li>2.2 Steps in planning</li> <li>2.3 Importance of planning</li> </ol> </li> <li>3. Decision making:             <ol style="list-style-type: none"> <li>3.1 Concept</li> <li>3.2 Types and process</li> </ol> </li> <li>4. Motivation:             <ol style="list-style-type: none"> <li>4.1 Meaning</li> <li>4.2 Theories of Motivation-Need Hierarchy                 <ul style="list-style-type: none"> <li>-Motivation-Hygiene Theory</li> </ul> </li> <li>4.3 Motivation through Employee Participation</li> </ol> </li> <li>5. Leadership:             <ol style="list-style-type: none"> <li>5.1 Meaning, Qualities of Effective Leader</li> <li>5.2 Leadership style</li> <li>5.3 Leadership Theories-Trait theory, Behavioral theory, Situational theory</li> <li>5.4 Emerging Leadership approaches-Transformational, Coaching, Entrepreneurial</li> </ol> </li> </ol> |
| <p>Practical Hours: 10</p>                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>• Case study</li> <li>• Group discussion</li> <li>• Sequence quizzes related to the subject</li> <li>• Project assignments</li> <li>• Q&amp;A sessions</li> <li>• Report writing</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <ol style="list-style-type: none"> <li>1. To provide guidance regarding the basic concept of accounting prevalent in organization</li> </ol>                                                                                                                                                                                      | <p><b>Introduction to Financial Accounting (6 Lecture Hours)</b></p> <ol style="list-style-type: none"> <li>1.1 Meaning and scope of accounting</li> <li>1.2 Accounting Transactions</li> <li>1.3 Capital and Revenue</li> <li>1.4 Balance sheets, P/L accounts and Cash Flow Statements</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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| 2. To provide guidance regarding ethical standards in IT<br>3. To help learners make various decisions that are in line with work setting values and following state and federal laws relating to IT | <b>Ethics for IT Professionals (6 Lecture Hours)</b><br>1.1 Concept of Ethics, Professional Ethics and its relevance<br>1.2 Technology for Betterment of Humanity<br>1.3 Code of ethics <ul style="list-style-type: none"> <li>• IEEE (Institute of Electrical and Electronics Engineers)</li> <li>• SANS (SysAdmin, Audit, Network and Security) Institute: The world's largest cyber security research and training organization</li> </ul> 1.4 Latest ICT Policy of Nepal and Electronic Transaction Act, 2063, IPR |
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### Major Teaching and Learning Strategy

- Lectures
- Case Analysis
- Project work
- Term paper
- Assignments
- Quizzes
- Presentation
- Guest Lecture Sessions in specific topics

### Assessment Plan

- In-semester: 50%**
- End-semester: 50%**

### References:

1. Griffin, R. W. Management. AITBS Publishers and Distributors, New Delhi
2. Robbins, S. P., Coulter, M. A., & Randel, A. (2021). *Management* (15<sup>th</sup> ed.). Pearson.
3. Hitt, M. A., Black, J. S., & Porter, L. W. (2011). *Management*. Pearson Education.
4. Driver, J. (2007). *Ethics: The Fundamentals*. Blackwell Publishing.
5. Ciulla, J. B., & Martin, C., & Solomon, R. C. (2018). *Honest work: A business ethics reader*. Oxford University Press.
6. YouTube tutorials and videos